RUMANIA

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DEVELOPMENT OF GERONTOLOGY

Research in the field of gerontology started as early as the nineteenth century. The outstanding neuropsychiatrist, Gh. Marinescu, was interested in the process of aging and the age pathology of the nervous system. He investigated particularly the colloid osmotic mechanism of aging and advanced an original theory on the subject (Marinescu, 1899, 1933).

C. I. Parhon, founder of the Rumanian school of endocrinology, investigated various aspects of the aging process, particularly the relationship between the endocrine glands and aging, other clinical and biological aspects of aging, and different prophylactic and therapeutic approaches to geriatrics (Parhon, 1955).

In 1949, C. I. Parhon chose me as co-worker in the field of gerontology and geriatrics. We both started a program of theoretical and clinical gerontological research in a home for the elderly. With a small number of collaborators, we achieved good results in the treatment of old age (Aslan, et al., 1950a, b). In 1952, the Institute of Geriatrics was founded at Budapest; its major objective was research on the problems of gerontology and geriatrics. I was appointed the director.

NATIONAL INSTITUTE OF GERONTOLOGY AND GERIATRICS

The institute was the first of its kind, so there was no organizational model available. Nevertheless, it was organized according to the three major directions of research in contemporary gerontology: clinical, experimental, and social. This structure has been recommended by the World Health Organization for similar institutes.

Because the achievements of the institute in the therapeutic, experimental, and medicosocial fields called for a more complex framework of organization and specialized and diversified research, it became the National Institute of Gerontology and Geriatrics (1973). The activity of the National Institute of Gerontology and Geriatrics has been directed toward the study of the aging of the human organism from the biological, clinical, therapeutic, psychological, and social points of view.

The better understanding of the mechanisms of aging of the human organism, the necessary steps for improving the health status of persons over 40 or 45, the maintenance of their work capacity, and the prolongation of the active life span have been major objectives of research at the institute.

For reaching these targets, the research activity at the institute has included biomedical and psychological fundamental and applied investigations on aging; geriatric curative and prophylactic steps; organization and guidance of other geriatric institutions of the sanitary network of our country; and appropriate gerontological training of the medical staff.

At present, the framework of organization of the institute contains various departments.

CLINICAL DEPARTMENTS

One department is for gerontological recovery where cross-sectional and longitudinal clinical and therapeutic studies have been carried out aiming at the development and improvement of the means of recovery and prolongation of the active life span. The target is to give a scientific basis to the improvement of the general health status to the maintenance and the prolongation of elderly's capacity. Research on the efficiency of eutrophic medication, age criteria, rhythm, and evolution of aging, as well as on the role of the genetic and endogenous determining factors, have been carried out.

In the clinical department for age pathology, the pathology of advanced age, as well as the clinical and therapeutic characteristics of chronic degenerative diseases, have been studied. Research on the interrelation of age-degenerative diseases and on the improvement of the prophylactic and curative steps in geriatrics has been carried out.

The clinical departments also include a number of specialized consulting rooms—cardiological, opthalmological, neuropsychiatric—and psychological and x-ray laboratories.

Research Departments

The department for the biology of aging includes laboratories specialized in clinical physiology, electrophysiology, hematology, biochemistry, bacteriology, immunology, cultures of cells, enzymology, drug conditioning, experimental biology, pathological anatomy, and medical electronics. This department has its own farm, which supplies animals for the experiments.

Research on subcellular, cellular, tissular, and systemic processes of aging have been carried out at this department through functional, morphological, and physiochemical explorations; pharmacological and pharmacokinetic studies; elaboration and conditioning of biotrophic products; findings of new methods of investigation and diagnosing; and studies of experimental gerontology. Research on the rhythms, pathogenesis, and pathology of aging, age criteria, and evaluation of therapeutic results have been carried out in close cooperation with the clinical departments.

The department of social gerontology includes gerohygiene, geropsychology, social biology, gerontology, methodology, social assistance, and gerodemography laboratories. The major objective of the activity at this department has been the study of the social, demographic, economic, medical, psychological, ecological, and cultural peculiarities of the process of aging, both of the individual as a member of a certain community or social stratum and of the social life as a whole.

From the methodological point of view, the department of social gerontology organizes and controls the research on old age prophylaxis and treatment, as well as providing the appropriate training of the medical staff.

Central Polyclinic

The elderly are attended by this polyclinic, which has specialized consulting rooms and laboratories. The polyclinic controls the geriatric, prophylactic, and curative assistance in different enterprises and other polyclinics in Bucharest, as well as the experimental centers for research and medical assistance. Prophylactic and therapeutic research on the efficiency of the biotrophic products on outpatients have been conducted. The central polyclinic also has consulting and treatment rooms for foreign outpatients.

Clinical Departments for Geriatric Treatment of Foreign Patients

Because a great number of foreigners have been interested in undergoing the treatment with ''Aslan's method'' (Gerovital H3), several clinical departments with specialized staff have been organized close to Bucharest and throughout the country. The Paro and Snagov departments of the Otopeniclinic function near Bucharest. Other clinics are located in Felix, Herculane, Eforie Nord, Neptun-Mangalia, Sinaia, and Caciulata.

RESEARCH

Therapy with Extracts and Procaine

Therapy of the aging has been considered with ever greater interest by both physicians and sociologists. The various treatments prescribed in the aging process are generally the result of conceptions regarding the pathogenetic mechanisms of aging. According to C. I. Parhon's conception, aging is a dystrophic state, which could be therapeutically influenced.

Aslan envisaged glandular treatments with thyroid extracts and, in certain cases, with thymus and epiphysis extracts, as well as treatments with extracts of ovaries or masculine hormones. Besides these therapeutic approaches, the procaine-based treatment Aslan started has been based on clinical and pharmacological investigations. The results obtained have pointed out the important part played by the nervous system in aging mechanisms.

Before 1946 this treatment was administered in trophic and circulation disorders, in asthma, arteritis, and syndromes for which it had already been mentioned in the literature (Leriche and Lafontaine, 1953; Visnevski and Visnevski, 1952).

From 1949 to 1951, Aslan experimented with intra-arterial therapy using procaine on young and old patients with circulatory and trophic disorders or with arthritis (Aslan et al., 1950a, b). The first reports mentioned the hair repigmentation, unknown at the time (Parhon and Aslan, 1953). Procaine-based therapy improved the memory and increased the muscular strength of elderly patients. This fact called for a systematic, long-term use of procaine-based treatment in old age.

Clinical long-term research started in May 1951 on 189 persons from the institute's home for the elderly. The patients were grouped for treatment with spleen and placenta tissue extracts, gland extracts of adrenal, pineal, and thyroid, vitamin E, yeast, and procaine. One group was used as control.

Procaine was administered in intramuscular injections three times weekly in a series of twelve injections a month with a ten-day interval between each series. Eight series were given annually, and the treatment was continued for a couple of years. After five years of comparative research, it was decided to continue only with procaine (Gerovital H3), vitamin E, and control groups. The elderly patients lived under similar environmental conditions and received the same food and care as the controls. The average age in the various experimental groups was 72 in the procaine-treated group, 75 in the vitamin E–treated group, and 73 in the control.

At first an ordinary solution of procaine was used. In 1951 Aslan elaborated the product; Gerovital H3 with procaine as the basic compound, the pH of which (3.3–3.5) provides a longer stabilization of the procaine molecule. Aslan, and later Gordon and Fudema (1965), pointed out that Gerovital H3 provides higher blood-procaine levels than procaine alone.

In order to evaluate the results of the procaine treatment in the old patients, Aslan has used a series of clinical and laboratory investigations in order to point out their biological and psychological age before and after the treatment. The following were the age criteria: the capacity to perform conditioned reflexes, the psychological reactivity, the velocity of the vascular reaction to various physical agents by plethysmographic and oscillometric tests, muscular strength shown by dynamometric tests, circulation time, and other biological aspects. The pattern of the patients' biological state was completed by biochemical investigations on the metabolic processes (concerning the proteins, fats, and hydrocarbonates) and hematological investigations (blood clotting, platelet aggregability, and blood morphological pattern among them). Life expectancy, morbidity, and morphopathological data were also studied.

After three years of treatment, the results obtained in the three groups were evaluated and compared. The best results were obtained in the group subjected to the procaine therapy; fewer improvements were noticed in the vitamin E group. A long-term treatment was then instituted for another thirteen years; mortality figures were recorded for the entire sixteen-year period. Mortality was 5.3 percent in the procaine-treated group, 10 percent in the vitamin E–treated group, and 16.5 percent in the control group.

From the beginning of the treatment with Gerovital H3, a favorable change in the patients' psychology appeared with regard to their life, activity, and family and social relations. Other changes were a diminution of the depressive state, a better mobility, and better articular elasticity (Aslan, 1956a, b, 1959a, b).

Gradually other scientific data have been added concerning the effects of Gerovital H3, including:

- Anabolic effect in 89.9 percent of the treated cases (Aslan, 1962; Aslan et al., 1965).
- Improvement of the psychic condition, memory, and time of fixation of conditioned reflexes, plus an increase in the number of sleeping hours (Aslan and Parhon, 1954; Parhon et al., 1955).
- Signs of reactivation of the regeneration process, noticeable in the skin and hair, and an acceleration of the consolidation time of accidental fractures (Crăuciun et al., 1959; Aslan et al., 1975).
- Improved circulation time (slowed in 81 percent of cases in 1952 but after eight years of treatment in only 23 percent of cases); increased vascular reactivity pointed out through oscillometric and plethysmographic tests; and increased muscular strength (Aslan, 1959a, b; Aslan, Vrăbiesou, 1955a, b, 1968).
- Modification of the proteinemy (the electrophoretic examination revealed a tendency of stabilization of the globulinic fractions) (Aslan and Parhon, 1954).
- Prevention of turning of micromolecules toward macromolecules, a general phenomenon in nontreated aged persons.
- Improved permeability of the capillary vessels for the various proteic fractions (electrophoretics) (Aslan et al., 1959).

The clinical results have been:

- Favorable effect in psychic and physical asthenia (Aslan and Parhon, 1954).
- Evident results in the ailments of the central nervous system induced by arteriosclerosis, in Parkinson's disease, pseudobulbar paralysis, cerebellar syndrome, and comitial crises (Aslan, 1956a, b; Aslan and Parhon, 1954; Aslan and David, 1960).

- Favorable effects in the treatment of sequela subsequent to cerebrovascular accidents and hemiplegia due to spasms, thromboses, or hemorrhages (Aslan and David, 1959).
- Improved motility, reflexes, trophic edema, and decreased spasticity (Aslan and David, 1957).
- Reappearance of estrogens and reactivation of androgens (Aslan, 1956a, b).
- Improvement in certain cases of the thyroid function and stimulation of cortico-suprarenal secretion (Thorn's test) (Parhon et al., 1956).
- Positive influences on all age-induced modifications of the skin, such as senile keratosis, cutis rhomboidelis and the troubles of hypothalamic origin, vitiligo, psoriasis, eczema, lichen planus, and alopecia (Aslan and Parhon, 1954; Aslan, 1957).
- Beneficial effects on scleroderma (investigations conducted on twenty cases for two to five years) and on collagenosis.
- Stimulation of hair growth and repigmentation (Parhon and Aslan, 1953).

The substance also proved to be quite active in the diseases of the circulatory system, particularly in peripheral and central atherosclerosis, in cerebrovascular disturbances, embolism, acrocyanosis, and trophic ulcers (Aslan and Parhon, 1954; Aslan and David, 1959; Aslan et al., 1970, 1971).

The influence of the substance on ulcers (gastroduodenal) was investigated in three hundred cases (Aslan et al., 1959). The diminution of disturbances was noticed after four or five injections (particularly when they were administered intravenously). There were also fewer digestive troubles such as hyperacidity and hypermotility, and the direct and indirect signs of ulcer were also influenced. Generally the results have been astonishing, particularly since they did not require any diet or a strict rest regime. Two old people ages 75 and 88 belonging to the group treated with procaine had bone fractures that were cured in a remarkably short time (forty days) with an endostal callus and complete functional recuperation.

Gerovital H3 is particularly effective in osteoarticular afflictions. More than seven thousand treated cases have proved the efficiency of the drug. The best results were noted in arthroses and spondyloses. The disappearance of the clinical subjective phenomena and the functional recovery were noticed in 28 percent of the cases, and general improvement was noticed in 61 percent. Only 11 percent of the cases were not significantly influenced (Aslan, 1956; Aslan et al., 1965; David et al., 1972).

The comparison of radiographies done before and after one to four years of treatment revealed the remineralization of the skeleton and the thickening of the bone bridges in 25 percent of the cases and the reestablishment of the subchondrial structures and the lessening of osteoporosis in 40 percent of the cases. Beneficial effects were also noticed in cases of rheumatoid arthritis, where small amounts of corticoids were added whenever necessary.

The stimulation of the defense reactions of the organism under Gerovital H3 treatment was described from a clinical and experimental point of view.

Aslan and colleagues (1965) noticed that during a severe influenza epidemic in the institute hostel in 1959, mortality in treated subjects was 2.7 percent whereas in nontreated ones it was 13.9 percent.

In research on rat peritoneal hystiomacrophage, Aslan et al. (1973) found the capacity of phagocytosis increased by 10.9 percent after treatment in vitro with Gerovital H3; and in the cells coming from treated animals, colloidopexy increased by 11.9 percent and phagocytosis by 14.4 percent as compared to the controls. They also followed the phagocytosis capacity of the human leukocytes of subjects submitted to a long treatment with Gerovital H3. The phagocytosis capacity was tested with inactivated Staphylococcus aureus. The average of the phagocytosis index for Staphylococcus aureus was 51.7 for the leukocytes from the controls and 68.4 in those treated with Gerovital H3, an increase of 16.7 percent.

Aslan et al (1977) directed research toward the possibility of controlling by Gerovital H3 the capacity of the lymphoid F_c receptors to bind immunoglobulin because of their importance in the mechanisms that concern the cell-mediated immune response. In rats treated with Gerovital H3, the percentage of F_c receptor bearing lymphocytes increased 44 percent in sixteen-month old rats, in comparison to the controls where the percentage was only 20. At the age of twenty-eight months, the percentages were 48 percent in the treated group and 13 percent in the controls.

Because the bonding capacity of lymphocytes decreases with age, our results show that treatment with Gerovital H3 leads to the maintenance of the lymphocytic reactivity that is the background of the cellular-mediated immune response. It is possible to assume, following these data, that long-term treatment with Gerovital H3 decreases the deterioration of structures, susceptible to becoming antigenic producers of autoantibodies. These data could explain the low level of autoantibodies in subjects treated with Gerovital H3, in comparison to the control group. Therefore Gerovital H3 appears to be a factor in the prevention of autoaggression in elderly subjects.

Investigations carried out at the Institute of Geriatrics on 150 elderly subjects treated with Gerovital H3 for nineteen years revealed a 36 percent increase of life expectancy (based on statistical tables for 1956–1957), but in controls, the calculated life expectancy exceeded the average life span 30 percent.

Mortality in the patients subjected to long-term treatment with Gerovital H3 was 4.1 percent; that in the control group was 16 percent. Crăciun and Tasoa conducted morphological studies particularly on tissues allowing the assessment of cellular metabolism; they pointed out the nuclear division in certain myocardial fibers. They concluded that this represents a reactivation at the level of the nucleus of the myocardic fibers correlated with the metabolic processes. The authors pointed out a morphological peculiarity: the absence of sclerosis consequent to atrophy of the liver, myocardium, brain, and endocrine glands.

Simultaneously with the clinical research, Aslan and her co-workers have conducted pharmacological investigations aimed at the mechanism of action of the procaine-based eutrophic medication. These investigations allowed a better understanding of the effect of this substance on the cellular biochemical activities, intermediate metabolisms, and integration systems of the organism. Stronger eutrophic effects have been noticed subsequent to the use of Gerovital H3.

The benefits derived from the regenerating effects of Gerovital H3 have been assessed based on experimental data. In one study the sciatic nerve was unilaterally crushed in rats. A group of rats received injections with Gerovital H3 (16mg/kg body weight) every two days, and the controls received only distilled water. Comparative studies were also carried out with Gerovital H3 and simple procaine (Aslan, 1962). The changes in the functional capacity of the triceps aural muscle were assessed through electromyographic determinations. Forty days after the pinching of the sciatic nerve, gravimetric determinations were made as well as the measurement of the transversal diameters of the muscular fibers of the injured area in comparison with the intact contralateral muscle.

A more rapid regression of the muscular atrophy was found in animals treated with Gerovital H3, the gravimetric difference between the muscle of the injured area and the intact muscle being 10.7 percent in treated animals and 22.6 percent in those belonging to the control group. Histologically the tranversal diameters of the muscular fibers of the injured area decreased by 2.3 percent in comparison with the normal condition in treated animals, whereas in the control group the decrease was 11.6 percent.

The effects of procaine with a pH of 7 on the regeneration processes has also been studied in comparative research. Thus the gravimetric difference between the muscle of the injured area and intact muscle were 19.6 percent, and the transversal diameters of the muscular fibers of the injured area decreased by 8.7 percent (in comparison with 10.7 percent and 2.3 percent in animals treated with procaine pH 3.3 Gerovital H3).

These investigations pointed out that Gerovital H3 has a stimulating action on the regeneration phenomenon at the level of peripheral nerves and of the striated muscles. Due to the synergism of procaine and that of potassium ions at the nervous and muscular level, it is understandable why Gerovital H3 (containing K) is more active than procaine alone.

In other studies experimental fractures at the level of the inferior third of the tibia and peroneum, without the mobilization of the respective limb, were used on forty female rats aged twelve to fourteen months, separated in two equal groups. The research aimed at studying bone regeneration. The study lasted sixty days. Radiographies performed every ten days revealed an earlier and a more intense periostal reaction in the Gerovital H3–treated animals (4 mg s.c. per day). Ten days later, the periostal reaction at the level of the fractured bones was present in 60 percent of the Gerovital H3–treated animals. The reaction was present in only 10 percent of the controls receiving distilled water. The precocity and intensity of the periostal reaction produced in those cases constitutes the essential stimulation for a more rapid consolidation of the two fractured bones. The results of the experimental investigations agree with the clinical pattern of the more rapid bone regeneration in Gerovital H3–treated elderly.

Stimulation of the tissues' regeneration by the biotrophic products based on procaine has been clearly shown in the research of Naum and Rusu (1973), who have established the quantitative variations of nucleic acids during hepatic regeneration. When it was shown that after twenty hours the rate of the DNA synthesis is much slower in old animals than in the young ones, it became obvious that in the treated old animals, the DNA synthesis is stimulated, and it approaches that of young animals. The stimulation of DNA synthesis determined by reduced doses of Gerovital H3 (0.4 percent) was also evidenced during research on secondary cultures of monkey kidney. The incorporation of labeled thymidine was 22.4 percent in treated cells and 6.2 percent in controls (Aslan et al., 1973).

Experimental research has also shown the anabolic effect of procaine, which was previously clinically observed. In Infusoria (Colpidium colpoda and Vorticella), we noted a stimulation of proliferation under the influence of procaine with low concentration solution, a fact indicating anabolic stimulation.

In 1950, investigations on rats drew attention to the anabolic influence of procaine, revealing weight increase and improved furry coat of the experimental animals (Aslan et al., 1950).

Aslan et al. (1965) have studied longevity in 1,840 rats. They pointed out a 21 percent increase of life span in Gerovital H3–treated animals. An improved general trophicity was also noticed, as well as an increased resistance to bronchial-pulmonary diseases. Fewer myocardial lesions were recorded on electrocardiograms (coronary irrigation disorders in 30 percent of the treated animals and 80 percent of the controls). Histological examination in the treated animals revealed a less important connective invasion in the myocardium and the striated muscle and less degenerative lesions in the renal tube.

Beside the anabolic action, a series of facts point to the vitamin-like biocatalytic role of procaine. Aslan has mentioned (Aslan, 1958, 1964, 1974; Aslan et al., 1965) Moeller and Schwartz's research (1948), which proves the stimulating action upon bacteria. This fact was also noticed by Kuhn (1948). They called the para-aminobenzoic acid a vitamin-like factor. Furthermore, the antisulphanylamide action of para-aminobenzoic acid has also been pointed out for hydroaminobenzoic acid, as well as for procaine (in the bacterial test, procaine can replace the para-aminobenzoic acid). The authors assert that carboxyl and amine groups are the prerequisites of the conservation of the vitamin-like effect.

Research we carried out (1960) in cooperation with Conniver upon the Staphylococcus aureus and Entamoeba coli have also confirmed the biocatalytic action of procaine.

The stimulation of the cellular proliferation under the influences of procaine and para-aminobenzoic acid has also been pointed out in research on Infusoria (Colpidium colpoda and Vorticella) by Parhon, Aslan, and Cosmovici (1957). Concentrations at which maximal stimulation had been obtained were 2 mg percent for APAB and 4 mg percent for Gerovital H3. The development of a certain yellow pigment in the Infusoria culture occurred in the presence of both substances, which proved their similar effects (Aslan, 1958). This yellow pigment has also been found by Mayer (1944) in experiments carried out with procaine and para-aminobenzoic acid on micobacterium tuberculosis.

Other research carried out on secondary cellular cultures from monkey kidneys showed that the concentration of 0.4 percent Gerovital H3 produced the stimulation of cellular proliferation. The average cellular density varied, depending on the age of the culture, between 218.9 cells/ml and 712.5 cells/ml in the Gerovital H3–treated group and between 163.5 and 480.3 cells/ml in the control group. The data point cut an intensification of cellular proliferation in the treated cultures. The average cell postmitotic life span was 72.4 days in cultures treated with Gerovital H3 having undergone fourteen passages and 62.3 days in nontreated cultures having undergone twelve passages.

Recent research carried out by Americans (Hrachovec, 1972; MacFarlane, 1974) supports the assertion that the effect of Gerovital H3 on the cellular biochemical activity has a much wider spectrum due to its capacity of inhibiting Monoaminoxidase (MAO). A correlation has been established between this effect and the procaine mechanisms of action.

The inhibition of monoaminoxidase was indirectly noticed by us in 1950, when we found that intravenous procaine injections administered to dogs produced an important increase of the blood pressure reaction toward epinephrine.

Procaine action at cellular level also involved restoring and maintaining the physiologic potential of the cellular membrane. The mechanism might be the densification of the cellular membrane—in Fleckenstein's opinion the "anelectronic repolarization" and the "stabilization" in Shanes'.

The same properties of the membrane have been pointed out by Teitel, Storesou, and Steflea (1965). Similar findings were pointed out in research on isolated organs of homeothermals and poikilothermals, the activity and reactivity of which toward certain substances are increased by procaine, thus providing a longer survival.

Considering the importance of activities on the organism as a whole, any improvements in its functional state brought about by the biotrophic treatment are particularly significant. The favorable influence upon depressive psychical states of the aged was also noticed (Aslan, 1956). The antidepressive effect of procaine has also been pointed out by Bucci and Saunders (1960), Siggelkov (1960), and other American researchers. Together with Parhon and colleagues (1955b), we found an improvement of memory, attention, and sleep in patients treated regularly with procaine.

Recently in a double-blind research using thirty elderly patients, Zung (1972) using placebo and imipramine, tried to evaluate the efficiency of Gerovital H3 therapy in depressive elderly patients. The clinical observations and psychometric tests performed before and after the four-week treatment pointed out the superiority of Gerovital H3 over imipramine.

Data recently published by American authors drew attention to the modification induced by aging and depressive states in the enzymatic activity of the nervous cell as well as upon the intervention of procaine at this level (Robinson et al., 1972; Hrachovec, 1972; MacFarlane, 1974). The increased MAO activity could play an important part in the biochemical modifications induced by aging and depressive states. As a matter of fact, depressive states have been correlated with the reduction of central amines (Bunney et al., 1967), which is due, as recently shown, to the increase of monoaminoxidase.

MacFarlane (1974) pointed out that the treatment of psychic depressive with irreversible MAO inhibitors (such as pargyline and phenelzine) is associated with severe adverse, sometimes lethal reactions. On the contrary, Gerovital H3, a reversible MAO inhibitor, may be safely used.

Although the mechanism through which procaine acts upon the nervous system is not known, the analysis of the effects it induces enables us to believe that it could play a part in the restoration of the electric potential and the biomedical substratum of the nervous cell.

Electroencephalographic research we carried out in 1963 with Brosteanu and Enachescu pointed out that people over 85 years subjected to long-term treatment with Gerovital H3 had, compared with the control group, a significantly higher (75 percent) alpha background activity, within normal limits for adults, a prompt reactivity to light, and no alternations to intermittent luminous stimuli. At the same time, the capacity for psychic effort increased, and the appearance of signs of fatigue was delayed.

Research has shown that procaine acts upon the nervous system not only at the whole molecule level but also through its hydrolysis products: diethylaminoethanol and para-aminobenzoic acid. The results obtained with the administration of procaine supports the assertion that diethylaminoethanol (resulting from the hydrolysis of procaine), through its transformation into diethylaminoethanol, is also a supplier of acetylcholine. In this way, procaine, through diethylaminoethanol, not only feeds the metabolic potential way, which determines the increased production of acetylcholine, but inhibits at the same time the hydrolytic action of acetylcholinesterasis; diethylaminoethanol is also a feeble antipsychotic.

In 1972, Hrachovec showed in research upon rat brain, liver, and heart that Gerovital H3 inhibited monoaminoxidase more intensely than hydrolytic procaine. According to MacFarlane (1972), the inhibition produced by Gerovital H3 is from 17.8 to 87.7 percent, the intensity depending on the dose.

All of these data, as well as the clinical studies pointing to the effects of Gerovital H3 on old people, support the assertion that the reduction of monoaminoxidase and its return to normal values affects positively the symptoms associated with aging. These are new proof of the action of procaine upon the central nervous system.

The research on clinical, biological, and social gerontology carried out at the institute have had direct applicability. The results obtained since 1950 in applying the biotrophic substances according to Aslan's method on two hundred thousand subjects proved its peculiar efficiency in the prophylaxis and therapy of premature aging. They aroused general interest and stimulated numerous studies, resulting in over six hundred published papers) both in our country and abroad. These achievements contributed to a large extent to the founding and development of the Rumanian school of gerontology, which has attracted hundreds of physicians and specialists from our country and from over eighty foreign countries to attend the specialized training at the institute.

Based on the results obtained, we started the prophylactic use of the biotrophic medication in 1954, when the first gerontological center was organized at an industrial enterprise in Ploiesti. Since then, the action has been extended within the numerous gerontological centers organized in the industrial and agricultural enterprises throughout the country. In collaboration with the department of social gerontology and the clinical departments, the gerontological center provide prophylactic assistance for persons aged 45 to 60.

The prophylactic orientation of the gerontological research, the main objective of which has been the prolongation of the active life span, is the characteristic of present and prospective activity at the institute.

SOCIAL GERONTOLOGY

The evolution of the demographic aging of the Rumanian population has been similar to that of the countries characterized by an intense economic-social development. Due to the rapid increase of the elderly population, the health care services have had to face new objectives, such as the founding of the department of social gerontology at the Institute of Geriatrics in 1958. The department has aimed at studying the medicosocial problems of the elderly and aged population. The main objectives of the multidisciplinary research carried out at this department have been studies on demography, anthropology, and social medicine; studies of applied medical and psychosocial sociology; medicosocial experiments; and methodological problems of the organization and functioning of new forms of geriatric assistance.

The research of social medicine had been focused on the main objectives in the present health care of the population: to decrease mortality and morbidity due to chronic degenerative diseases and to prevent premature and pathological aging through studies aimed at pointing out the causes of the peculiar rhythm of aging in certain geographical areas.

Longevity and environmental conditions have been studied based on medicosocial inquiries and psychological tests on fifteen thousand persons aged 85 and over (40 percent of the total of longevous persons in Rumania). The results of the study have pointed to the higher biological vigor and to the increased resistance to different diseases; 44 percent of the longevous persons had direct and collateral longevous ancestors.

The original contribution of the Rumanian research in social gerontology has been the assessment of the biological aging of certain groups of population; the criterion of average biological age has been used as a synthetic, positive indicator of the health status of the elderly population. This methodology has been applied in research conducted on the population suffering from endemic goiter in certain geographical areas in our country; it revealed the peculiarities of premature aging due to the thyroid endemic dystrophy specific to these groups of population. Research conducted on populations in those geographical areas with particularly increased frequency of longevity pointed out the peculiarities of the delayed process of aging in the population.

Both the theoretical and practical multidisciplinary research conducted by the department of social gerontology have enabled a better understanding as well as the solution of certain problems specific to the medicosocial assistance of the elderly population.

Medical care has been concerned with problems of prophylaxis, in centers specializing in the prevention of age pathology within the sanitary network including the gerontological centers and the dispensarization of the aged population, and with curative functions through outpatient and hospital care (geriatric departments in general hospitals, and homes for the chronically ill aged persons).

Social assistance has focused on services for those elderly living in the family environment and the institutionalization of the elderly (in homes for the elderly and for retired persons).

Medical units with prophylactic profile include the gerontological center, functioning within the medical units in enterprises, which provides prophylactic medical assistance for the working staff aged over 45. The center also provides medical supervision for elderly workers through semiannual medical check-ups; the use of the prophylactic therapy with biotrophic substances; the solution of social problems raised by changes in working place, labor protection, and conflicts; and the study of work and age interaction.

The main objective of the gerontological center has been prevention of ailments through periodic medical examinations, prolongation of the active life span, and the maintenance of working capacity. Also it assessed certain objective indicators concerning the evolution of health status, such as body weight, height, thoracic perimeter, pulse, blood pressure, and so on, as well as social indicators concerning the environmental conditions of the working staff, the age when they started to work, their occupational career, the characteristics of the working place and dwelling, nutrition, energetic and neuropsychic effort pointing out age-induced functional modifications, and the part played by one's occupation in the process of aging.

The analysis of the results obtained at the 144 gerontological centers between 1961 and 1970 point out the following:

- The necessity of specialized medicosocial assistance for retired persons.
- The necessity of preventing certain chronic diseases or detecting them at the onset.
- A 38 percent decrease in the number of sick leave days.
- A prolongation of the active life span pointed out by the fact that 24 percent of the persons served by the centers continued to work after reaching the retirement age.
- The maintenance or the increase of the work capacity in 41 percent of the working staff as a result of having to achieve a certain quota.

There are gerontological centers at present in 219 enterprises. By 1980 coverage of the active elderly persons by centers will be generalized throughout the country. Care for the elderly population has been differentiated by age subgroups. Biomedical peculiarities and the morphophysiological status of this group of population have been taken into consideration. Persons over 60 are cared for at the territorial polyclinics, where specialized consulting rooms such as internal medicine, cardiology, and neuropsychiatry have been at their disposal. They have been subjected to complete medical investigations every six months, and the results have been listed in the personal medical records.

Care for nearly 25,000 elderly persons achieved the following:

- Complex medical examinations, which pointed out the onset of chronic diseases of which the patient had no knowledge (such as cardiovascular diseases and diabetes).
- The active supervision of the patients, enabling the prevention of complications.
- The use of specialized therapy, as well as that of different geriatric recovery techniques and treatments with biotrophic substances.
- Certain social ergotherapeutic steps.

The action thus has a dynamic character, and the medical and social steps taken at the proper time ensure its quality and efficacy.

The dispensarization was applied also to persons aged 90 and over because of their high percentage of death from acute diseases. The main objective of the monthly supervision of these persons has been the prevention of the onset of the acute pathology, different complications, and worsening of certain chronic diseases. This step led to a 56 percent decrease in the number of deaths in longevous persons as compared to the year previous to the dispensarization by preventing acute diseases (influenza, bronchopneumopathies, acute afflictions of the digestive or renal apparatus). Research on this age group also includes a study on the interrelation between the environmental factors (physical and social) and longevity, and studies of data concerning life, dwelling and working conditions, peculiarities of nutrition, family life, the psychological aspects concerning personal nature, temper, affectivity, and so on.

Curative medical assistance includes outpatient care at the medical territorial units and polyclinics and geriatric units in polyclinics with specialized consulting rooms. Seventy geriatric consulting rooms function at present; by 1980 the number will increase to 130. Geriatric consulting rooms also provide outpatient treatment, including recovery and biotrophic therapy. The main aims of the geriatric consulting rooms have been specialized medical assistance, complex medical investigations specific to the age pathology, and geriatric training of the medical staff.

The medical dispensary with its gerontological profile functions within the retired people's associations. The retired medical and sanitary staff give consultations and treatments to members of the association. In cooperation with the territorial medical units, it provides treatments in different resorts and refers patients to specialized consulting rooms.

The organizers intend to provide medical assistance at home with the help of the dispensaries for patients unable to get around because of advanced age or chronic diease. Hospital assistance has been provided by the hospital for adults, but it has been inconvenient, particularly in cases of acute diseases. The pathology of the aged population, characterized by chronic degenerative diseases simultaneous with a deficient background, requires special units, apparatus, and conditions.

Hospital assistance within the units with geriatrics profile include geriatric departments, such as those functioning at the NIGG. Their main obejctives have been to diagnose disease, making use of the investigations required by the specific age pathology; to apply special treatments aimed at recovery (the physiotherapeutic, kinesiotherapeutic, and mechanotherapeutic techniques and medical gymnastics have been thus developed); to administer associated medications with biotrophic substances (Gerovital H3 and Aslavital); to develop appropriate diet required by the treatment; to supervise the convalescence of patients; and to meet the social needs of the aged through the social services.

Social assistance for elderly includes a series of social programs. NIGG has organized an experimental unit, referred to as the village for the elderly, for those from rural areas. This unit includes eighty rural houses with two or three adequately furnished rooms, vegetable gardens, and orchards, where the elderly can work according to their ability. The village is at the center of a rural community, so the elderly are in permanent contact with different age groups.

Among the multidisciplinary research conducted by the department of social gerontology, mention should be made of the study of the process of demographic aging and its medicosocial consequences; research on premature and pathological aging of the population living under environmental conditions with increased risk factors, such as geographical areas with endemic population or increased radioactivity; the study of the influence of the psychic factors on premature and pathological aging; research on the use of the elderly's working capacity; the study of the adaptive processes of the aged organism in relation to the changes of the main parameters of the environment, such as pollution, inappropriate diet, obesity, and working conditions; the study of modifications in the social status of the elderly due to retirement; studies of the mutations within the family structure and relations of the elderly; and studies of the dwelling, educational, and cultural needs of the elderly.

The results of the research conducted by the department of social gerontology have been published by specialized journals and in the *Treatise on Social Gerontology*.

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